ActInf Textbook Group ~ Cohort 3 ~ Meeting 1 (Welcome and Onboarding)

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Hello everyone, welcome to the Active Inference Par et al textbook group.

Welcome to Cohort 3.

Our journey begins here.

It is January 25th, 2023.

And this is an onboarding and overview meeting.

So as with all meetings, it is being recorded.

We know that no single meeting time will accommodate every single person's schedule every time.

So live meetings of this type are recorded.

And then later in this day, like 10 hours from now or something, there's the office hours, optional as everything is in Discord.

And these meetings are recorded.

They'll be available here.

So for example, you can see from Cohort 2, there will be like a link for the video recording.

This first meeting is onboarding and overview.

So the first part is going to be going over the textbook group and the structure of the coda, which I know looks large, but it is possible to focus in and steward the parts that you're interested in and not the parts that you're not.

So I'm going to give an overview of the structure of the textbook group in terms of synchronous and asynchronous participation.

And in the Zoom chat, please feel free to ask any questions you want just so that we can address them.

The goal here is to reduce your uncertainty, increase your confidence around participating.

There are so many ways to participate and we want to welcome and accommodate and encourage everyone coming from totally different learning preferences and backgrounds.

So I'm starting on the very top left of our coda and the link has been put in the chat and also note that I'm using dark mode.

So coda might look like that, but I'm using it in dark mode.

Welcome to the Active Inference Institute.

Many people here in this room right now, I know have been fundamental and with us for years.

Others, this might be one of your first times joining.

That's all great.

And I'll put this link in the chat.

The textbook group is one of many activities that are ongoing at the Institute.

So here we are in the Paradol textbook group and there's other education and research oriented activities that you can explore and find out more about.

But for now, we're going to be focused on the textbook group and we'll reach out to other sources and places as we need.

The textbook itself is open source.

We are talking about the textbook from 2022, the Active Inference textbook by Parr, Pizzullo, and Friston.

And it's an open source textbook, which is awesome.

That allows us to facilitate us, of course, working in this way.

And also it allows you to download the textbook.

So you can download the PDF right here.

And I know there's some other efforts that some of you have started involving like making it a living textbook.

The directory page, and you can hide and show the sidebar.

Sorry, one second.

The directory page has information on different sections of the Coda and what you can find there.

Live meeting page summarizes when our live meetings are happening.

And we're specifically in cohort three.

So here we are.

This list is just auto populated from the names that people have put.

So let us know if you want anything to be changed.

Just active inference at gmail.com.

And also this is a field that you can edit if you want to add your desired contact information.

It wasn't added for anyone.

But if you want to put contact me on this way at this time, or if you're interested in that, like you can add that here for connecting within the cohort.

And then there's that cohort three live meetings page that has the date and time of all of our meetings.

And that's where the video recordings are going to be placed as well.

So broadly, today we're here to go on an overview, take any introductory thoughts and questions, and also hear from each other.

Like, what are we excited to learn and what is going to motivate us in this optional but extremely exciting and useful quest that we're on?

So, then we will have two weeks per chapter.

The chapters are not super long.

So two weeks is a good amount of time to give reading and or listening to it a chance.

Take notes, and we'll talk more about what that can look like and be.

But on the meeting for a given chapter, we're going to focus on that chapter.

And sometimes, again, we'll reach into other resources, but we're going to focus our attention on the focal chapter.

And then to a slightly lesser extent, other parts of the textbook, because people might have a question that, like,

Oh, that's going to come up in chapter seven, or that's going to come up in chapter three.

And then this textbook is not the only star in the constellation of emerging active inference literature.

And so there's going to be many times where we'll reference, like, other papers, other programs,

and especially we'll reference a lot of the live streams, the several hundred paper-oriented and model-oriented live streams that we've done at the Institute.

And so many of the critical papers and researchers we've highlighted in these streams.

So it's like a really awesome resource.

Okay.

As for onboarding and participation.

So these are all evolving documents and in an evolving open science, creative commons space.

So with our participation in synchronous and asynchronous institute affordances, like this Zoom room and this Coda,

we're agreeing and abiding by rules of general respect and also of the open source status of what we're collaborating on here.

And we are students in the textbook group.

We're focused on learning.

It's going to lead many of us towards project ideas and collaborations, which we're also very excited to facilitate.

We expect and prefer and hope you to be an engaged learner.

However, there may be multiple types and levels of novelty or frictions.

So communicate early, communicate often.

Let us know how we can improve things for you.

There's a lot of opportunities and bandwidth to help you literally specifically with what you want.

So make it known.

And hopefully with this very capable and diverse group, we can make it happen.

We've also included some students practices.

You can just click these sections away if you're not needing them or not interested in them in a moment.

But students practices, like what is it to really learn active inference?

What are we really doing from a verb perspective?

And so here are some suggestions and some options ranging from annotating, writing, systematic reading, having leisure time, making programming, constructions, adding questions, adding ideas and insights, adding projects.

There's many ways that we can participate and be involved.

And if you're finding something that's exciting and engaging for you and you feel like it's adding value to your active inference learning, like let's add it.

Other learners may be interested in it too.

So that's overall what participation is like.

This is the pace or the rhythm that we'll be following in the sessions that are weekly at this time.

We're going to get in the next three months through the first half of the book, which is like a lot of the conceptual and theoretical underpinnings.

And then the second half of the book is about a recipe for modeling itself.

So you can see like cohort two, who meets actually an hour before this.

If you ever want to join, welcome to.

Same link.

From September to November 22, we went through chapter one through five.

And now we're going through chapter six or 10.

And so we'll do something similar, like from May to August, we'll go through chapter six or 10.

Again, everybody can join everything, go through multiple times in the beginning or however it's going to work out in your life and practice.

But we're going to be having this pace with the recorded video meetings going through chapters one through five in the coming months.

And then if that is all you have bandwidth for, that's great.

However, to engage with the material on your own time is really where the learning is going to happen.

And some of your edits and contributions on your own time will be in this coda as well in sections that we'll look at.

Others will be in your own documents.

So join when you can.

Watch the video recordings when you can.

Comment early.

Comment often.

Let us know how we can help and improve things.

So let's go into first just any generals questions on this.

And then we'll go into like the textbook itself and how we use coda around the textbook.

All right.

In the textbook section itself, this textbook is the first active inference textbook.

And it's the first version of the first active inference textbook.

There's a lot of other critical work, like vital work as well as skeptical work.

And Carl Fristen, who is one of the central researchers in this field, and Paul and Pazulo, two younger researchers who have worked with Fristen,

have created this first version of a first textbook published at the end of March 2022.

So about eight months ago.

It's a big moment for our field.

And you here are all some of the very first to be going through this textbook, which is positioned, as Fristen has said,

towards a master's student in computational neuropsychiatry looking to apply this to their own research.

That may literally describe some of you.

Others may have quite a different context.

Doesn't mean it's going to be not fit for you.

In fact, it's been exciting to see how many different backgrounds and people have resonated with this textbook and with, of course, all the related work.

So this is like a flag in the ground for active inference.

But also, this isn't a one-stop shop.

So anywhere that you're like, I wish they would have explained this or what is the meaning of that?

Like, that is what we're doing in this group.

Importantly, this textbook doesn't have any questions.

There's no assessment or basic comprehension questions.

And that is also a huge aspect of what our work is here.

So to go through just some of these pages and how we use the coda, there are some equations in this book you might have noticed.

How do we plan to understand and work with these equations, recognizing everyone's different capacities and preferences for equations?

Well, one thing that we do is for every equation, and this work has been done by previous cohorts.

So now we're in the, like, improve phase, not the first construction phase.

For every equation, we have a screenshot and a verbal description.

So anywhere in the coda, someone can be like, I'm curious about equation at 2.5.

Mouse over it?

What is it?

Oh, okay.

Okay.

Equation 2.5.

Cool.

Well, but what does it mean?

Oh, the variational free energy is etc.

So here's a natural language description of these equations.

I'm just muting people who are making a little voice.

Thank you.

So you'll gain familiarity with coda and how everything looks.

But when you see a canvas that you can pop out, you can find a natural language description.

And it's not there for every single one.

So these are, like, the exact kind of contributions that people can make that have extremely high leverage.

Anywhere that you can add a description or add comments or notes, add it in LaTeX, those are, like, the kinds of local improvements that, in our group context, with, like, 30-plus future cohorts of us here, can be, like, actually transformational.

I mean, equation 2.5 is one of the essential equations.

How can we come to understand it?

Well, we can have discussion.

We can ask questions on it.

We can do, like, what Jakob had previously done, like, show how the lines of the equation are related to each other.

So there's an incredible amount of work that we can do around improving the equations.

And coda facilitates us using every row as something that can be referred to.

So that's equations.

We have figures.

Also, every figure has been brought in.

Not everything is fully tagged or fully described, but figure 2.2.

Oh, there it is.

So we can use it and reference to it as, like, something stable and as something that we can ask questions about and so on.

Boxes and tables, all of those have been brought in.

In the code page, this may or may not be relevant for the first half of the book for you, depending on where you're at with code.

But we have a lot of active inference code implementations.

And we can go more into this as people want to.

But there are a lot of code-based implementations, especially in Python, but also in MATLAB and other languages for people who want to, like, see what it looks like in a program form.

And something that we'll come back to again and again and just always a nexus to develop is we are going to be talking about math, natural language, graphical visualizations and art, programming, philosophy.

Like, we're going to have a lot of different things coming together.

There's no single path within active inference or to active inference.

People have different backgrounds and familiarities with different areas.

And so maybe for one person, it's like the philosophical implications that are exciting for you.

And then you'll come to see the programming.

Maybe somebody else sees the programming adjacency, and then they'll come to be excited about the philosophy or about how to visualize it.

So we'll be moving across those domains.

And yes, there is a facet of daunting scope.

We're here in a new context, and this is an opportunity to learn the way that we want to learn, and that is learning actively.

In chapters, notes, and questions, we have, like, some sections that if you want to zoom in on a specific chapter.

So here's, like, the table of contents, and then here's the fine scale table of contents with the sections.

So somebody can say, like, I'm talking about section at 1.4.2.

Again, maybe that's, like, relevant and you want to do that, but we do reference different sections, and then that makes it, oh, okay, that's what this section's about.

Here's what page it's on.

Within a given chapter, like chapter one, there is a subset of questions that have been asked on chapter one, and then also there's just other discourse on these pages.

And so we have those for all the chapters and for the appendices.

That's the textbook section.

It's primarily just focused on stripping out and reformatting the material of the textbook so that it can be referenced and enriched to this extent.

Like, everywhere that equation 2.5 is used, it's augmented by what we've added to this table.

So mark once, use many times.

In the discourse on the textbook, this is, like, really where the creativity and the generativity of our cohorts is going to shine.

We want to facilitate active learning, and a huge element of active learning is this question-based perspective.

So here's an overall questions table, and also there's a filtered view that will only be for our cohort.

I believe someone has already added a bunch of questions for chapter one, which is awesome.

Like, the questions are shown on the left, and you can just add by hitting enter and just adding a new question down here.

And then also you can pop up the answers in discourse.

So here, and also I'm using control and plus or minus or control and scroll to make Coda bigger or smaller.

That can be quite useful.

So these questions that people have asked range from basic and testing comprehension of even very yes or no or multiple choice,

all the way to really the kind of connections and relationships that are the substance of having fluency and comprehensive understanding of active inference

on through speculative philosophy questions and open areas of research.

If each one of us here added like one question per week and refined zero or a few questions of discourse,

we would have a truly transformationally different active inference learning environment worldwide.

And I don't think that's an overstatement.

There are not many groups working on this textbook, especially in this collaborative capacity.

The textbook doesn't have any questions associated with it.

And we're very close with the authors.

We're going to have opportunities to use what we're developing here broadly and play a role in the future of active inference education.

That's one of the most important reasons why adding questions and refining questions is so critical.

Like we've all heard, oh, if you have the question and X number of other people in this room or lecture might as well.

That is and will be happening globally.

When we use the active inference ontology, which I'll come to after showing the other sections,

we also can leverage our questions in the discourse on them.

And in the live meetings, what we will tend to do is focus on questions that haven't been addressed yet that have been upvoted as interesting.

Like we'll just pop it open and let people share.

And we'll all open up this and we'll start taking down some thoughts and questions.

And we're just like getting it on the page.

Being active, making contributions.

It doesn't need to be unified or consistent in the initial framing.

But these are the kinds of things that are going to help people in their synchronous and asynchronous active learning for this textbook and even beyond.

Because a lot of these questions are, of course, more general.

So ideas and insights is like if you just want to lock in an idea or an insight you had and just crystallize that so that others can see that.

Oh, chapter one.

I was like, oh, here's what this means.

Might be shocking for some.

Might be incomprehensible for some.

It's all good.

Just when you add in here, it becomes part of our collective space.

So I highlight questions a lot because that is one of the most important ways that we can increment our understanding and go from that like moment of friction or uncertainty.

Like what does that mean?

Or why doesn't they define this?

Or what is this equation doing here?

Or what does that figure mean?

It's like in that moment of not understanding, make the choice to write down the question and our group will have an exponential capacity.

But if everybody just wonders about it and nobody writes a question or augments the question, then it's going to be like we're all moving ahead linearly.

The math learning group.

So this is active to the extent that people want it to be.

It was a suggestion and brought forward by some of the participants in the first cohort.

And in this little subsection, which again, put it aside if you don't want to.

We really wanted to focus on the foundational understanding of the math.

That's one of the stickiest parts of this field.

And it is also fundamental.

Like the relationship between surprise, uncertainty, and hidden states.

That relationship isn't a poem.

It's beautiful.

But also there is mathematics that underlies these topics.

And we want it to be accessible and rigorous for all.

So in this section, which again, if somebody wants to like steward this math learning group and bring together basic resources on matrix algebra, on stochastic systems, on calculus, or wants to write summaries like some have done that come through the sections.

Those are super useful.

Yana, any thoughts on being stuck or not having questions?

It's not required to ask questions.

So if the expression that you have in response to the material is not question oriented, if it doesn't pose itself as a question, it's all good.

Or as a question every time, it's all good.

And so what is it that affords the emergence of questions?

Whatever it is, when it emerges, feel free to capture it.

If it emerges as something different or even something ineffable or non-active in that moment, it's also all good.

Add any other comments or feel free to add things.

But like number one imperative is to stay in the game, stay engaged.

And if we live to play another day, we advance our knowledge of active inference.

And just for like a little context, this is something that would be taught in a graduate school optional seminar.

For some of you, that might literally describe you.

But for those who are stepping into something that from a legacy perspective isn't what they would be stepping into,

like they're, at least I personally, but I also hope more broadly that the bravery of that could be understood.

In project ideas, participants have added various project ideas.

So contact these people or like add your own ideas, anything from doing audiobook recordings to specific modeling,

different educational possibilities, mathematical derivations.

So improving like on one hand, the rigor, while we're also working on the other hand with accessibility,

keeping those integrated resources.

It may become relevant and Ali contributed this initial table.

It may become relevant to refer to other resources.

So you'll find a lot of other materials here.

And in future textbook groups, there's like a feedback form,

which we'll update at the end so that we can give our feedback.

And then also like if you want to share the link for people who want to join future cohorts

and also just add like other feedback, like what can a voluntary global textbook group be?

How can we learn and apply active inference?

Those are some of our questions.

Errata are a few little typos and not just like conceptual uncertainties, but what appear to be typos in the book.

And also I'll note that on 17 UTC on February 27th,

Thomas Parr will be joining for live stream, first author of the textbook.

So in the next month, we will like highlight certain questions that we want to surface when Thomas joins.

It'll be like in a Zoom just like this.

Of course, it'll be recorded.

Also, if anybody like wants to join, we can talk about this in the coming weeks.

But this is really awesome and unique opportunity to ask Thomas directly certain questions that we have.

So that should be awesome.

Okay, last section.

And then we'll go into more open group discussion.

This active inference tables section has a few tables that are going to be very useful.

This is the active inference ontology.

The active inference ontology is a central project since the earliest days of ActInf Institute.

And one can see a little bit more about it at this public site, but there's also more to it.

The ontology at this point consists of definitions and translations for a reduced set of 74 core terms.

And a slightly broader list of supplemental terms.

Terms that are maybe not vital, like as prerequisites for understanding ActInf, but they're things that we might mention, like when we're learning.

And this is, again, the public ontology site.

So here we have natural language definitions or different terms.

Just like a dictionary uses natural language terms to describe other natural language terms, that's what we're doing in the active ontology.

And just like being fluent in a language means that you can recognize and use the language, understand what the entities and the terms are, the topics, and how they're related and why,

So this is a set of terms where if one is fluent in perceiving and acting upon them, one has fluidity in active inference.

Also, the ontology has translations.

So we have translations into Spanish, Portuguese, French, Italian, Czech, Russian, Hebrew, Persian, Chinese, Japanese, limited in Hindi and Turkish.

If you want to improve one of these ontology language translations, or you want to add another language, that is like an incredible service.

Because, because, let's think back to that equation 2.5, when we see this blue text, these are ontology terms.

So the natural language definition can pop up.

And when we want to type something, we can do like, at generative model.

We can then translate into other languages very effectively and have integrity across representations and languages, including mathematics and non-mathematics, but also different natural languages.

So, the ontology is how we're taking a systems engineering approach to learning and applying active inference.

As this incipient ecosystem of learning and application is really, truly just beginning, we want it to censor accessibility and rigor from the beginning.

And people are going to use the ontology in different ways, but having it as a resource is already been vital.

And for those who see this as like a possible way to work, as a way to make improvements to, for example, translations and definitions, it's a big option.

And we have a table here so that we can reference it throughout, because as you'll see, like questions use, where possible, ontology tags.

We're talking about minimizing surprise.

Now, there still is a lot of complexity around what surprise is, but like, this is a starting point.

This is like something clickable and interactable that we can always go deeper into.

So, last tables, then we'll go into the question section.

So, here's the live streams table, or you can use the link here.

Here are the 200 plus live streams that we've done.

I'll just highlight like live streams are paper oriented.

So, we've done 53 papers.

So, here you'll find a huge number of relevant papers, and if you could look up like eco.

Okay, here's some discussions on ecology.

Live stream are paper oriented.

Guest streams include a broad range of guests in act, inf, and adjacent areas.

And like, always, please feel free to suggest anyone or make an introduction if you want to see something highlighted.

Model streams focus on models and applications of active inference.

Like, for example, the Python implementation of PyMDP, which we'll talk about.

And then, down on the bottom, you'll see the Par Pizzulo Friston textbook.

And here, you'll see cohort one's videos.

So, if you just feel like watching cohort one's videos, then you can see like, here's their playlist going through.

And here's just all the students with their optional contact information.

So, again, just to close the section, and then the last 20 minutes, we'll have anything we want to ask or discuss.

So, overview provides some information on active textbook group.

Has information on our live meetings.

Textbook is mainly oriented around focusing discussion on a given chapter and on the contents of the textbook.

Textbook is mainly oriented around focusing on a particular topic.

In discourse, we have questions, ideas, and insights, and project ideas.

These are some places to make contributions to.

And then, active inference tables just hold some reference tables.

Okay.

I'll look at the questions in the chat.

And then, anyone who wants to raise their hand or add something else in the chat.

Okay.

If I plan to engage asynchronously and have questions on a chapter ahead of the cohort three schedule, where should I ask them?

Great question.

So, you can ask them also in the questions table.

Here's cohort three questions.

So, they're all here.

If you have a question on chapter three, four, five, just add it here.

We'll get to it when we will get to it.

So, hope that addresses it.

But, you know, you're not limited to asking only what we're going to discuss that week.

If you have something on chapter three, just add it right here.

Thanks, Everett, for answering it.

I was concerned the question wouldn't promote discussion until the group made it to that chapter.

It can.

We're not hyper only focusing on only the text of the chapter, but that is the group regime of attention.

So, it will facilitate discussion when people do get there on their own time or in the group.

And you say, I plan to implement some of the active principles rather soon.

By being here, we are already all implementing the principles and anywhere that you can make your contribution.

Like, we're all going to be exploring and whittling away at this forest in our own paths.

So, just add it and it'll come into play.

Hope that addresses it.

Lucas, and then anyone else?

Oh, yeah.

I know you mentioned something about listening to the book.

I did take a look at it and it looks like there's a lot of mathematical equations and graphs.

Like, would you say that in order to have the best experience, it should be in a physical

copy to read rather than to listen to Unaudible or something?

Great question.

A few ways to talk about that.

Like, the audio version of the first chapter was very easy.

The introduction in the first chapter, which is what I made because they didn't have equations.

The natural language descriptions of the equations, one can imagine, will be vital for the audiobook.

Because one can then say, okay, equation 2.5.

And then read the equation like this.

We don't exactly have that yet.

And, like, depending on how deeply one wants to go into the math,

like, the zeroth level is like, they just put math here.

There's some other section here.

Carrying on.

The next level might be just like, okay, this is about free energy on some things.

What are those things?

Well, okay, you can go one level deeper.
Okay, what is that function?
Or what is that functional?
Okay, one level deeper.
So, we're all, like, you can go down the rabbit hole your whole life on 2 plus 2.
And on the other side, you can dance with equations like this.
Even if you don't have a formal mathematical background,
we're not generating these equations.
We're not par.
Or at all.
We're just understanding what, to some extent, they put.
And then we ask questions about, like, what the role is of these equations
and how they're used and everything like that.
So, listening may be a good experience.
But share your experience if you found listening to it to be, like, useful
or found something lacking.
I'd like to find a way to articulate mathematical equations audibly.
Yeah, thank you.
Like, that is exactly what this is about.
And that can't be done in even any automated way.
Like, this, and it's not the only way.
So, if someone has another idea about how to audibilize equations,
that's also very cool and creative.
But also, like, for those who feel like they have the capacity
to just take a first stab, and you can select the text
and then just write, like, you know, write a comment.
I'm not sure.
That's all good.
But it's an example of how, like, one contribution
can leverage people in other language communities
who are in different stages of learning and applying.
Anyone else, feel free to, like, raise your hand.
Or just go for it.
So, just while people are typing any questions
or thinking if they want to raise their hand,
we'll have two weeks per chapter.
Each time we have a discussion, we'll, like, go to that page.
And we'll first assess if we have any questions

that people, like, want to discuss.
And then, often in the first discussion
or at the end of the previous,
of the second discussion of the previous section,
we'll have kind of just, like, a look ahead
where we look at what the section is.
I'll give a quick overview on the structure of the textbook.
So, this could be seen at chapter notes and questions,
or it can be seen in the table of contents.
The book is separated into two parts.
In the next three months, in this cohort three,
we're going to be focusing on the first five chapters.
The first chapter is an overview.
It's non-technical, has a lot of conceptual materials,
but there are no equations or formalities.
Chapters two and three describe the low road
and the high road to active inference.
It's a framing that we will all come to know and love.
The low road is kind of like the how
in terms of using the mechanics of Bayesian statistics.
And the high road is like the ultimate why.
This is like the imperative of persistence
and the free energy principle.
And the two roads meet in the middle
with active inference,
which is using Bayesian statistics
to model and implement that persistence imperative
from the free energy principle.
All roads lead to active.
That's chapters two and three.
Chapter four, having described the two roads
to active inference,
is the heart of active inference,
which is about modeling.
And it's about this specific type of model
called a generative model.
The generative model is like the heart
or the kernel of active inference.

And we'll get there.
Chapter five then turns to one of the areas
where active inference has been most characterized
and studied,
which is in mammalian neurobiology.
And chapter five goes through a few neural systems
of mammals and talks about active inference modeling
in those settings,
and then provides a table
that has citations to many other places
that active inference has been applied.
So that's the conceptual half of the book
is overview.
What's the big picture?
What's the positioning?
What are we talking about with act-inf?
What are the two roads to get there?
Or what two roads meet in act-inf?
What is the heart of act-inf
in terms of generative modeling?
And then what is one of the most critical areas
that act-inf has been applied,
which is mammalian neurobiology.
The second half,
which is going to be starting in a few months
for this cohort,
or people can show up one hour earlier
and check in with cohort two,
who's working on the second half now.
The second half starts
with a informal recipe
for designing active inference models.
Kind of like a step-by-step,
but it's like a process
or a recipe for this kind of modeling.
So like,
we're going to be like,
where's the modeling?

Where's the modeling?
It's like,
it's coming.
And it's now for people who want to do it.
Then chapter seven and eight
describe a few different
important subtypes of generative models.
Chapter nine describes using data
in generative modeling.
And chapter 10
is kind of like the opposite of chapter one
or the other end of the bookshelf.
It's like summarizing
and giving an overview.
There are some appendices
that may be of value to you,
but you may dive into those
as they're referenced in the text
or no worries if not.
So that's kind of the broad picture
and the pacing
that we're going to go through.
Kate, thanks for the awesome question.
Yes, if anyone has like Python resources,
putting them into code
or resources could be awesome.
Here's like a bunch of implementations
that we have.
And in the recent
PyMDP stream,
I'll put this link in the chat.
This in the video description,
here's a Google Colab notebook
that doesn't need any tuning.
You can just run this in your browser
and it is doing active inference.
So if you want help with understanding
like the Python basics

and the syntax of Python,
I'm sure many people can share resources about that.
But like this notebook
is awesome
for actually working through it
and playing with different parameters.
And in the first half of the textbook,
we're building a lot of the conceptual
and formal understanding
so that when we're in the programming,
we'll understand what these matrices are.
And there isn't just one path.
Maybe somebody wants to just work through this
and then come to the formalisms later
or work on the philosophy
and then come to the math.
It's like there's just a million paths.
So as long as we're in the game and active,
we're making it happen.
Okay.
Any other thoughts or questions?
Okay.
Just again, raise your hand
or put it in the chat.
Let's just do a little tiny look ahead.
Next two weeks,
we're going to be discussing chapter one.
The preface
is written by Carl Friston.
It's just two pages long
and it includes this
absolutely endearing confession
that Carl was not allowed to.
It was not in affordance
for Carl to write this textbook.
Those who know may be familiar
that his writing style is very erudite.
And so the book,

it was not the desired style.
So that's just very nice by Carl.
But the preface doesn't really
have any big pieces.
Chapter one is the overview.
Yes.
Thanks for those,
like all different time zones
coming through.
Hopefully like the office hours
at the different time,
the recordings.
And of course,
like if you want to suggest
another time that you could commit to,
it may be possible
to also add that.
But also,
the synchronous activities
are totally optional
and one's own reading
or listening contributions
are primarily on their own time.
Chapter one is an overview.
It's introducing the main questions
that active inference
is going to seek to address.
That is the target
that they are painting
for this textbook
is active inference
as a unified principle
for mind,
brain,
and behavior.
And how can organisms persist
amidst environmental exchange?
That question

is the heart
of section 1.2.
And we see
a really
simplified version
of an action perception loop
that we're going to
be focusing on.
Agent,
world,
generative model,
agent,
generative process,
world,
niche.
Doesn't only apply
to mammals
and continents.
It turns out
that this
formalization
of perception,
cognition,
action,
and impact
is going to be
immensely
transferable
and composable.
Active inference
is sometimes
described
as a first
principles
perspective
and
it'll be awesome
to hear more

discussions
on what people
think that means.
But one way
to think about it
is like
this is kind of
a first principles
approach
in terms of
a how
with Bayes statistics
and this
is like
a first
principles
why
in terms
of
persistence
as an
imperative.
There's the
introduction
of two
attitudes
towards science,
NEATs
and Scruffies.
NEATs are
seeking unification,
Scruffies
embrace
heterogeneity.
And so
do we want
to have
a unique

biological model?
Let's have a
journal for
every synapse.
Let's have
a different
university
for every
single
follicle.
Like
at some
point
we have
composition
and
harmonization
not
homogenization
of the
diversity
of biology.
And so
that neat
scruffy
dialectic
guides a
lot of
these
discussions
about
particulars
and
generals.
The structure
of the book.
there's two
parts to

the book.
These are
aimed at
readers who
want to
understand
active
inference.
First part
that's us
and those
who seek
to use
it for
their own
research.
Second part
also that
is us.
We will
get through
both.
The first
part is
going to
introduce it
conceptually
and formally.
It's going
to refer to
some other
alternate
theories of
cognition.
And so
just to
plant the
seed.

When people
are asking
questions or
even
skepticisms
or critiques
of active
inference,
you can
ask yourself
or them,
is this
like a
critique of
modeling
in general?
Like,
would any
usage of
any model
fall victim
to that
criticism?
Which doesn't
mean it's
not valid.
It just
means that
it's like
an immensely
far-ranging
critique.
Or is
there a
unique
critique that
is being
made of

what we're
actually
learning
here?
And it's
not always
immediately
clear.
The goal
of the first
part is to
provide a
comprehensive,
formal,
and self-contained
introduction to
active inference,
its main
constructs and
implications for
the study of
brain and
cognition.
Well,
we'll see
if it is
comprehensive,
formal,
and self-contained
and where
we have
questions,
we can
ask.
Where we
want to
add,
we can

add.
It's just
the first
version of
the first
textbook that's
been in the
niche for
eight months,
and we can
be part of
that feedback
loop and
make it
happen.
Part one,
active inference
and theory.
That's the
first five
chapters.
This is
going to
give some
overviews on
key terms
that are
going to
come into
play.
Why are
we talking
about
surprise?
Why are
we talking
about
adaptive

control?
Why are
we talking
about the
action
perception
loop?
Here's
that figure,
the two
roads to
active
inference.
High
road,
starting
on the
top
right,
free
energy
principle.
Low
road,
starting
on the
bottom
left,
Bayes
theorem.
Description
of the
high road
and low
road,
which are
going to
be chapters

two and
three.
Chapters
two and
three.
Chapter
two,
low
road.
Chapter
three,
high
road.
four,
active
inference
formally.
That's
where we're
going to
get to
the
generative
model.
What is
the
generative
model?
What is
generative
modeling?
That is
the heart
of active
inference.
The
linear
model is

at the heart of linear modeling, and the generative model is going to be at the heart of active inference modeling. And then chapter five is going to move from the generative model in the abstract to a setting where generative models have been applied at this point the most densely in literature, which is mammalian neuroscience. Here are

some of the
key differences
between
active and
alternate
frameworks.
The
unity of
perception
and action
in terms
of a
shared
imperative
to
minimize
free
energy.
The
accommodation
of
planning
and
policy
selection
censoring
sensory
data.
Diverse
cognitive
operations
attention,
memory,
anticipation
conceptualized
as inference
over
generative

models
consistent
with the
Bayesian
brain
hypothesis.
Perception
and learning
as active
processes.
Action
as goal
oriented,
cybernetics
and the
biological
plausibility
of different
constructs.
Part two,
active
inference
in practice.
Check out
that, but
we're not
going to
focus on
it
immediately.
And
1.5,
this chapter
is just
summarizing
the first
half of
this book

previewing
the
perspective
that will
be
unpacked.
The book
is divided
into two
parts,
understand
and use.
They hope
it will
persuade
readers that
active
inference
offers not
only a
unifying
principle
under which
to understand
behavior,
but also
a tractable
approach to
studying
action and
perception
in autonomous
systems.
so it's
about 14
pages long.
We have
two weeks

to discuss
it, so
we'll be
able to
surface
many
reflections
and sensitivities
that people
are having
to
chapter
one.
That
concludes
our
first
meeting.
Following
this meeting,
the recording
will be posted
as an
unlisted
YouTube
video here.
Actually,
immediately
following this
meeting over
in the
Discord,
which I'll
copy the
link in,
we're just
going to
hang out

and have
education open
hours.
It's kind
of convenient
if you want
to continue
the discussion
on the
textbook or
learning or
education.
Thank you
all for
joining.
I will
close the
recording and
looking forward
to seeing
you next
week.
Thank you.
Bye.